Transmission Lines

The following subjects will be covered

- Wave propagation (guiding) on a TL.
- Analysis and examples.

Transmission Line Equations (Summary)

Transmission line Equations

$$V(z) = V_o^+ e^{-\gamma z} + V_o^- e^{\gamma z}$$

$$I(z) = \frac{1}{Z_o}\left(V_o^+ e^{-\gamma z} - V_o^- e^{\gamma z}\right)$$

Characteristic Impedance

$$Z_o = \frac{R' + j\omega L'}{\gamma} = \sqrt{\frac{R' + j\omega L'}{G' + j\omega C'}} \qquad Z_o = R_o + jX_o$$

Propagation Coefficient

$$\gamma^2 = (R' + j\omega L')(G' + j\omega C') \qquad \gamma = \alpha + j\beta \qquad \alpha = Re\{\gamma\} \qquad \beta = Im\{\gamma\}$$

Attenuation Coefficient

Phase Coefficient

Transmission Line Examples

Lossless Line

$$G' = 0$$

$$R' = 0$$

$$\gamma = \sqrt{(j\omega L')(j\omega C')}$$

$$\alpha = 0$$

$$\beta = \omega\sqrt{L'C'} = \omega\sqrt{\mu\varepsilon}$$

$$Z_o = \sqrt{\frac{L'}{C'}}$$

$$v = c/\sqrt{\varepsilon_r}$$

$$\lambda = 2\pi/\beta$$

Distortion-less Line

Attenuation is frequency-independent
Phase is linearly dependent of frequency

$$\frac{R'}{L'} = \frac{G'}{C'}$$

$$\gamma = \sqrt{R'G'}\left(1 + \frac{j\omega C'}{G'}\right)$$

$$\alpha = \sqrt{R'G'}$$

$$\beta = \omega\sqrt{L'C'} = \omega\sqrt{\mu\varepsilon}$$

$$Z_o = \sqrt{\frac{R'}{G'}} = \sqrt{\frac{L'}{C'}}$$

Examples

1) Design a 50-Ohm air line at 700 MHz with phase constant of 20 rad/m.

$$\frac{\beta}{Z_o} = \omega C' = \frac{20}{50} \quad \Rightarrow \quad C' = \frac{20}{50\,\omega} = 90.9 \;\; pF/m$$

$$L' = {Z_o}^2\, C' = 227 \;\; nH/m$$

2) Find the circuit elements of a 60-Ohm distortionless line with attenuation constant of 20 mNp/m and wave speed of $0.6c$.

$$\alpha Z_o = \sqrt{R'G'}\sqrt{\frac{R'}{G'}} = R' \quad \Rightarrow \quad R' = 1.2 \;\; \Omega/m$$

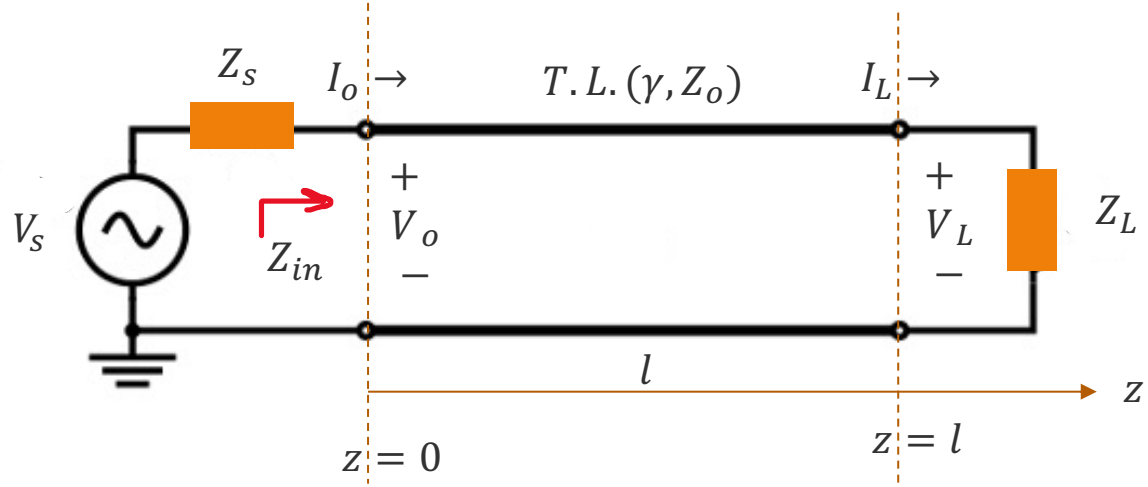
$$\frac{Z_o}{v} = \frac{\sqrt{\frac{L'}{C'}}}{1/\sqrt{L'C'}} = L' \quad \Rightarrow \quad L' = 333 \;\; nH/m$$

$$\frac{\alpha}{Z_o} = \frac{\sqrt{R'G'}}{\sqrt{\frac{R'}{G'}}} = G' \quad \Rightarrow \quad G' = 0.33 \;\; mS/m$$

$$C' = 92.6 \;\; pF/m$$

Analysis of Transmission Line Circuits

Consider the following general circuit with the following axis set-up



$$V(z) = V_o^+ e^{-\gamma z} + V_o^- e^{\gamma z}$$

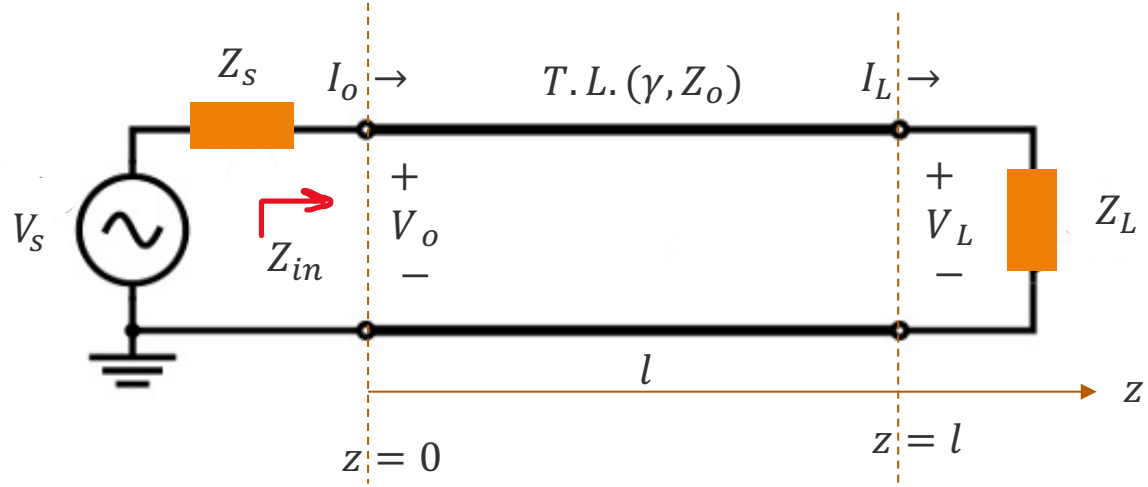
$$I(z) = \frac{1}{Z_o}\left(V_o^+ e^{-\gamma z} - V_o^- e^{\gamma z}\right)$$

To find the two unknowns (V_o^+ and V_o^-), boundary conditions are needed.

1) From the input (source) side.
2) From the output (load) side.

Analysis of Transmission Line Circuits

Consider the following general circuit with the following axis set-up



If input voltage and current are known, then

$$V(z=0) = V_o^+ + V_o^- = V_o$$

$$I(z=0) = \frac{1}{Z_o}(V_o^+ - V_o^-) = I_o$$

Solve for the unknowns

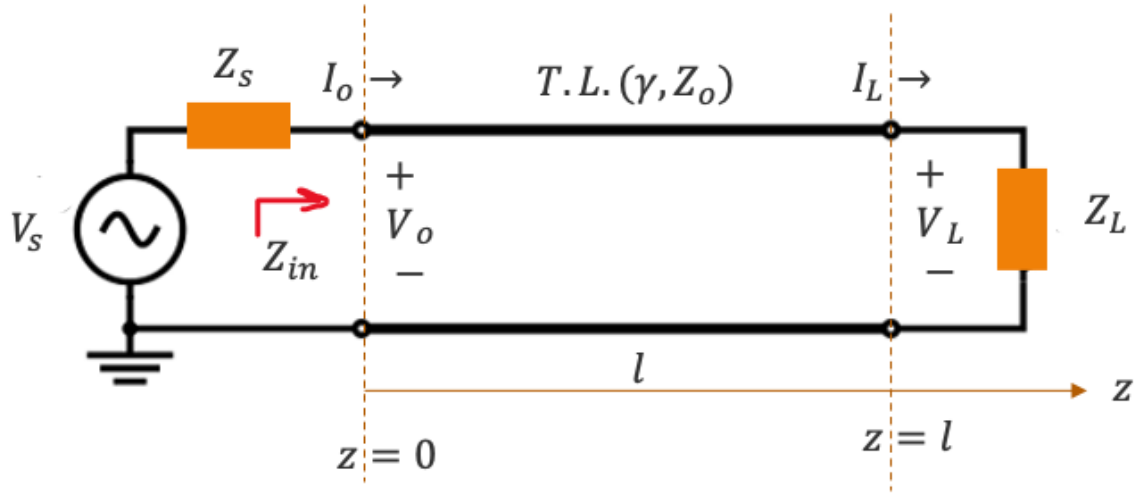
$$V_o^+ = \frac{1}{2}(V_o + Z_o I_o)$$

$$V_o^- = \frac{1}{2}(V_o - Z_o I_o)$$

Note that: $V_o = V_s \dfrac{Z_{in}}{Z_{in} + Z_s}$ $I_o = \dfrac{V_s}{Z_{in} + Z_s}$

Analysis of Transmission Line Circuits

Consider the following general circuit with the following axis set-up



If output voltage and current are known, then

$$V(z=l) = V_o^+ e^{-\gamma l} + V_o^- e^{\gamma l} = V_L$$

$$I(z=l) = \frac{1}{Z_o}\left(V_o^+ e^{-\gamma l} - V_o^- e^{\gamma l}\right) = I_L$$

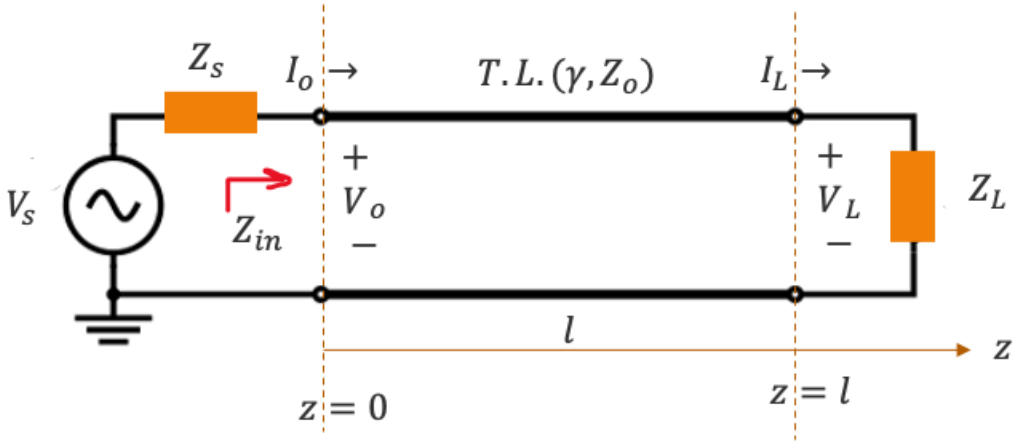
Solve for the unknowns

$$V_o^+ = \frac{1}{2}(V_L + Z_o I_L)\ e^{\gamma l}$$

$$V_o^- = \frac{1}{2}(V_L - Z_o I_L)\ e^{-\gamma l}$$

Analysis of Transmission Line Circuits

Consider the following general circuit with the following axis set-up



To evaluate the input impedance, Z_{in}

$$Z_{in} = Z(0) = \frac{V(0)}{I(0)} = \frac{V_o}{I_o} = \frac{V_o^+ + V_o^-}{\frac{1}{Z_o}(V_o^+ - V_o^-)}$$

$$Z_{in} = Z_o\left[\frac{Z_L + Z_o \tanh(\gamma l)}{Z_o + Z_L \tanh(\gamma l)}\right]$$

Lossy line

$$Z_{in} = Z_o\left[\frac{Z_L + jZ_o \tan(\beta l)}{Z_o + jZ_L \tan(\beta l)}\right]$$

Lossless line

Example

A 2-m-long transmission line operating at 10^6 rad/s has the following properties α=8 dB/m, β=1 rad/m and Z_o=60+j40 Ohms. The line is connected to a source with V_s=10∠0 V and Z_s=40 Ohms. The connected load is Z_L=20+j50 Ohms.
Find Z_{in}, I_o and $I(L/2)$.

$$\alpha = \frac{8}{8.686} = 0.921\ Np/m \quad \Rightarrow \quad \gamma = 0.921 + j \quad (1/m)$$

$$\gamma L = 2(0.921 + j) \qquad \tanh \gamma L = 1.033 - j0.03929 \quad \Rightarrow \quad Z_{in} = 60.25 + j\ 38.79\ Ohms$$

$$I_o = \frac{V_s}{Z_s + Z_{in}} = \frac{10\angle 0}{40+60.25+j\ 38.79} = 93.03 \angle -21.75° \quad (\text{mA})$$

$$V_o = I_o Z_{in} = 6.667 \angle 11.62° \quad (\text{V})$$

$$V_o^+ = \frac{1}{2}(V_o + Z_o\ I_o) = 6.687 \angle 12.08° \quad (\text{V})$$

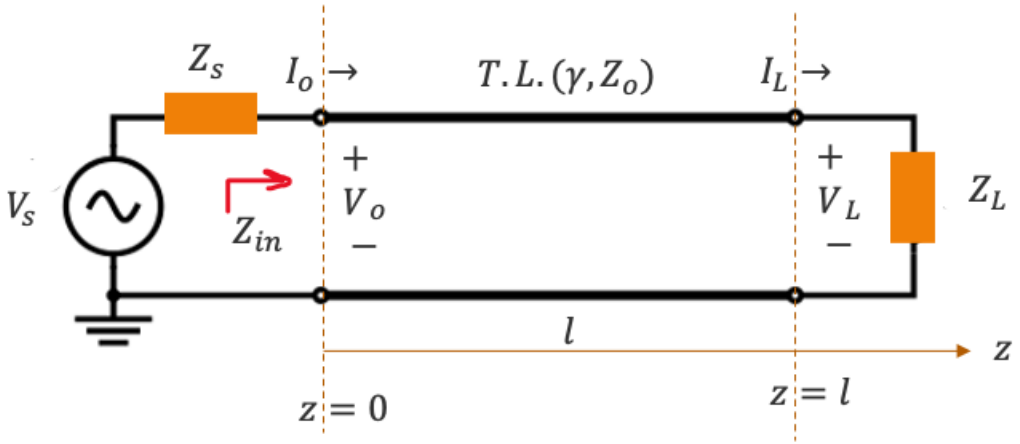
$$V_o^- = \frac{1}{2}(V_o - Z_o\ I_o) = 0.0518 \angle 260° \quad (\text{V})$$

$$I(z = L/2) = \frac{V_o^+}{Z_o} e^{-\gamma L/2} - \frac{V_o^-}{Z_o} e^{\gamma L/2}$$

$$I(z = L/2) = 35.1 \angle 281° \quad (\text{mA})$$

Analysis of Transmission Line Circuits

Consider the following general circuit with the following axis set-up



To evaluate the voltage reflection $\Gamma(z) = \dfrac{V_o^- \, e^{\gamma z}}{V_o^+ \, e^{-\gamma z}}$ $\Gamma(0) = \dfrac{V_o^-}{V_o^+}$

At load $(z = l)$ $\Gamma_L = \dfrac{Z_L - Z_o}{Z_L + Z_o}$ For a "matched" load $(Z_L = Z_o)$ $\Gamma_L = 0$

At any location $\Gamma(z) = \Gamma_L \, e^{-2\gamma z}$

In general, loads are complex $\Gamma = |\Gamma| \, e^{j\theta_\Gamma}$

Q & A

Send me an email if you have any questions.